



4357 - Interpreting JWST surveys with spatially-resolved mock line emission spectra from cosmological hyper-refinement simulations

Cycle: 2, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

The unprecedented infrared sensitivity and spatial resolution of the James Webb Space Telescope (JWST) is revolutionizing the study of galaxies and supermassive black holes (SMBHs) in Active Galactic Nuclei (AGN) from the nearby universe to the earliest stages of galaxy formation. However, maximizing the scientific return of JWST observations requires high-resolution, physically predictive models of the co-evolution of galaxies and SMBHs and careful development of realistic mock JWST observables from simulations for direct apples-to-apples comparison to galaxy surveys. We propose to leverage recent advances in (1) cosmological hyper-refinement simulation techniques, (2) SMBH growth and feedback algorithms, and (3) stellar and ISM physics in the FIRE-3 galaxy formation model, along with (4) state-of-the-art chemistry and 3D line radiative transfer solvers to produce synthetic spectral Integral Field Unit (IFU) data cubes with unprecedented information content. This will enable detailed interpretation of NIRSpec IFU and MIRI MRS spectral cubes for a large suite of emission line tracers from the mid-infrared to rest-frame UV for surveys of galaxies and quasars from the nearby to the distant universe. We will directly connect emission line tracers with the intrinsic physical conditions of gas, test standard reduction pipelines to infer physical properties from observed IFU data, and create a public data repository of synthetic spectral cubes.

OBSERVING DESCRIPTION

This proposal responds to a need for detailed spatially-resolved synthetic spectra of high-resolution galaxy formation simulations with physically predictive models for the multi-phase structure of the ISM, the impact of stellar feedback, and the growth and impact of SMBHs. Our proposed methodology combines some of the most advanced methods available for modeling galaxy evolution from cosmological initial conditions and for converting simulation outputs into realistic synthetic IFU observations. We anticipate that the results of this project will greatly benefit the interpretation of NIRSpec IFU and MIRI MRS spectral cubes and increase the scientific output JWST extragalactic surveys.